

Super-Resolution AR: Enhanced Image Visibility for AR Imagery

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ABSTRACT

In AR applications, there may be situations in which the visual target is not clearly visible/legible because it is too far or small. The unclear part of the imagery can be captured and magnified, but the image quality can still be problematic with the aliasing artifacts by the limited resolution. This poster proposes to apply deep learning-based upscaling to enhance the low-resolution images. We developed a prototype system that can capture an image and upscale/present it to the user. The pilot study demonstrated that upscaled imagery improved image clarity, the ability to find hidden information more quickly, and user experience.

1 INTRODUCTION

Augmented Reality (AR) combines the real world and computer-generated content by 3D registration. AR scenes can be viewed using a video see-through or optical see-through (OST) display both of which are equipped with built-in cameras. In either case, there may be situations in which the visual target is not clearly visible/legible because it is too far or relatively small. In such a case, the unclear part of the imagery can be captured and magnified, but the image quality can still be problematic with the aliasing artifacts by the limited resolution. That is, distant real world objects captured by the built-in camera and the virtual objects (e.g., image texture) may not have sufficient resolution for viewing them up close (see Figure 1).

One solution therefore is to simply employ a very high resolution camera as was proposed in [3,5], which is costly/redundant (or adds weight to the headset/glass), and cannot deal with virtual imagery created originally with low resolution. In this poster, we propose to apply the upscaling technique to enhance the low-resolution image target without additional hardware. Upscaling is the process of recovering high-frequency details from a low-resolution (LR) input provided to increase the image size to create a super-resolution (SR) image [2]. In addition, the upscaling process enhances the clarity of image contours and boundaries, thus contributing to a more refined visual experience.

In particular, OST displays (e.g., AR glasses) display AR objects in a semi-transparent manner (e.g., due to limitations in the optics and insufficiently relayed light), thereby introducing inherent visibility challenges in addition to the aforementioned resolution issue [1]. Presenting upscaled SR imagery should help overcome such a difficulty and enhance the overall augmented reality user experience. We developed a prototype system that can capture an image in a given AR scene (using the built-in camera from an AR glass), and upscale/present it to the user. Through a small pilot study, we validate the projected effectiveness.

2 PROTOTYPE IMPLEMENTATION

A prototype system according to the following usage scenario was developed. A user wearing an AR glass (Microsoft HoloLens 2)

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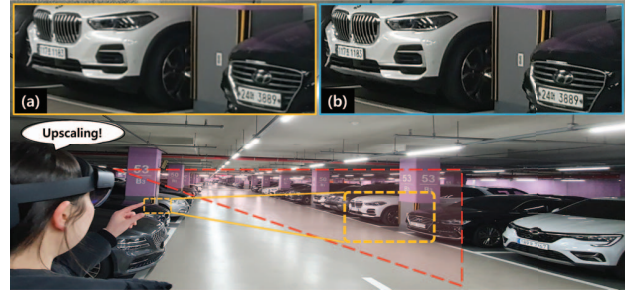


Figure 1: Overview of a prototype AR system used to capture a part of the image in a given AR scene (yellow box (a)), upscale it to (b), and display it to the user-worn AR device.

would indicate an intent to observe the part of the given AR scene in more detail by speech input, and designate the region with the hand gesture (see Figure 1). The region is captured and subsequently sent to the server for upscaling, and the result is returned for display.

The AR application was developed with the Unity 3D game engine (version 2021.3.22f1) and the HoloLens Mixed Reality Toolkit. Note that the built-in camera of the HoloLens has a resolution of 1920×1080 pixels. The communication between the server and the AR application was carried out. For now, the system does not run in real-time. The turn-around time takes about 40 seconds (about 19 seconds to upscale it and the rest for other operations such as communication latency, capture, and display).

As for the upscaling, we chose a deep learning-based SR model called the MZSR architecture with a Direct x2 pre-trained model [4] that combines the strengths of meta-learning and zero-shot approaches. The meta-learning employs the pre-trained parameters and allows for a rapid adaptation of the target image, while the zero-shot based approach also helps the upscaling performance by training internal information to handle previously unseen image types.

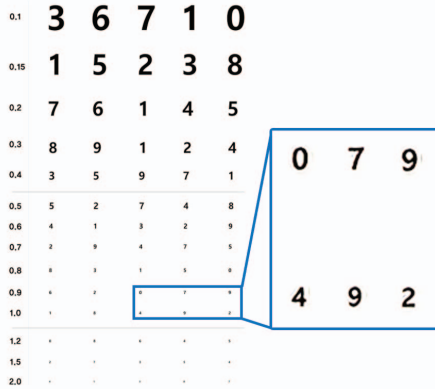
3 EXPERIMENT AND RESULTS

We conducted a pilot study asking the user to carry out a visual task with the original low resolution (as-is) or imagery upscaled by the implemented prototype. Thus, the study was designed as a 1×2 within-subjects. The experiment was run with two distinct visual tasks: (1) reading off letters/numbers from an eye examination chart and (2) searching/identifying numbers from a complex scene imagery (Hidden number game). 12 participants (8 males, 4 females, aged 20 – 34, $M = 26$, $SD = 4.69$) were recruited from the university.

3.1 Eye Examination

The eye examination task was employed as it is one common way to assess vision and the ability to focus on and discern objects. The participant was instructed to wear the HoloLens and stand 3 meters away from a white wall onto which the eye chart was augmented. These distances ensure that the image is entirely visible within the AR device's field of view.

The presented order of treatment (original or upscaled image) is counter-balanced. If participants could not read more than three letters/numbers at the same level of visual acuity (see Figure 2 (a)), we recorded the values and then asked them to rate the clarity of the image provided to them on a 7-point Likert scale. Once two treatments were completed, the task was concluded.



(a) Task 1: Eye examination. The left 5 columns are the original eye chart image, and the blue box represents a part of the upscaled eye chart image.



(b) Task 2: Hidden number game¹. The figures on the left are the original, and the figures on the right are the upscaled images. Note that only one image was presented for each treatment during the task.

Figure 2: The pilot evaluation study was conducted with two visual identification tasks: (a) reading off the eye examination chart and (b) searching and identifying hidden numbers from 1 and 40.

Given the experimental design and limited samples, the Wilcoxon signed-rank test was employed with a 5% significance level. The results are shown in Figure 3 (a) and (b). Unlike the expectation, there was a slight decrease in visibility observed in upscale images, even though it was not statistically significant (the level of visual clarity ($z = -1.02$; $p = .30$) and visual acuity ($z = -0.54$; $p = .58$)). The letters/digits in the chart were in black with a white background with maximum contrast. As such, the upscaling system interpolated the pixel values at the boundaries using a midpoint value of 0.5. Thus, it is possible that the originally sharp text may have actually gotten distorted with artifacts, leading to negative results.

3.2 Hidden Number Game

In this second task, the objective was to find and correctly identify numbers from 1 to 40 that are concealed within a complex image and pattern (see Figure 2 (b)). We presumed that the upscaling process could improve the visibility of the image and task performance.

The procedures closely resembled the first task. Participants were directed to read the concealed numbers for three minutes. Subsequently, they assessed the level of visual clarity on a 7-point Likert scale. The presented images/conditions were counter-balanced.

¹<https://absolutist.com/online/mysteriez-hidden-numbers/>

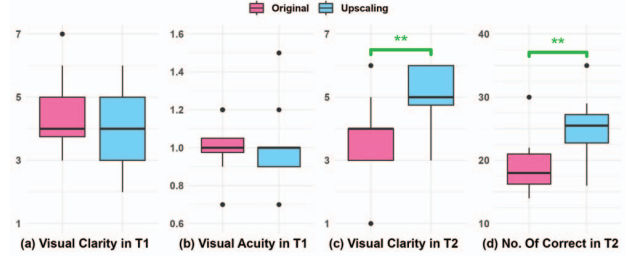


Figure 3: Comparison of the visual clarity and acuity in the first task (T1), and visual clarity and number of correct in the second task (T2). Significantly better effects were found for T2 (** $p < .01$).

Unlike the first task and as expected, the upscaled image showed significantly better visual clarity compared to the original image ($z = 2.67$, $p = .007^{**}$), as shown in Figure 3 (c). The images utilized in the Task 2 featured a variety of colors and intricate patterns, allowing the upscaling to exhibit its original effectiveness. Given the dynamic nature of the real world, we considered the results of the Task 2 to be more valuable than the Task 1 in terms of the applicability.

The number of correct answers as a task performance was evaluated by the Wilcoxon signed-rank test with a 5% significance level. Figure 3 (d) shows that participants found significantly more hidden numbers correctly in the upscaled condition compared to the original ($z = 2.79$, $p = .006^{**}$). The findings clearly indicate that participants were able to find more numbers more quickly in the upscaled condition compared to the original.

In summary, the pilot study did validate the projected effectiveness of the upscaling to improving visibility and interaction in AR. In addition, we also learned that considering how the deep learning models generally processes the image to upscale it, the results can depend on how the foreground target is situated for the background.

4 CONCLUSION

In this poster, we introduced a prototype system capable of upscaling AR images by applying the deep learning based SR model. The pilot study showed the potential for significant improvement in visibility by applying the upscaling technique. However, extended processing and communication delays caused significant usability issues as well. In future research, the attainment of real-time performance for AR interactions will be explored.

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